Image-Based Visual Servoing for High-Speed Multicopter Interception via Safe Reinforcement Learning

A Staged Curriculum from Kinematic Validation to Control-Barrier-Function Safety with In-Policy Projection

Project Master Technical Report

Compiled June 1, 2026

Abstract

We present an end-to-end study of training a reinforcement-learning policy for image-based visual servoing (IBVS) interception of a maneuvering aerial target, progressing through a nine-step curriculum from a kinematic pipeline-validation stage to a full six-degree-of-freedom (6-DOF) multicopter with realistic perception, environmental wind, and control-barrier-function (CBF) safety enforcement. The contribution is twofold. First, a *reproducible staged methodology*: each stage isolates a single new source of difficulty (action semantics, observation realism, sensor noise/delay, attitude dynamics, safety, disturbance), warm-started from the previous stage, with deterministic 200-episode evaluation at a fixed seed. Second, a *rigorous safe-RL ablation*: we compare an external CBF-QP safety filter against an in-policy differentiable CBF projection ("HardNet"), and through a controlled multi-seed study (variance baseline $\to$ entropy/curriculum interventions $\to$ auxiliary feasibility loss $\to$ λ ablation) we show that the in-policy projection, once its gradients are properly conditioned, attains the best worst-case robustness (36.7% success under $\sim$92% detection dropout and 1.5 m/s wind) while preserving nominal performance ($\sim$91%) and hard attitude-safety guarantees ($\leq$ 0.02% limit-exceedance across thousands of episodes). Every numeric result is taken from committed evaluation logs.

Contents

1 Introduction and Problem Formulation

1.1 Motivation

High-speed interception of an agile aerial target using only on-board vision is a canonical safety-critical control problem: the interceptor must keep the target inside its camera field of view (FOV) while closing distance rapidly, all under partial observability, sensor noise, and actuator limits. We adopt the image-based visual-servoing (IBVS) formulation of Yang et al. (arXiv:2404.08296), in which feedback is defined directly on the image plane, and we learn the guidance policy with Proximal Policy Optimization (PPO).

1.2 Coordinate Conventions and Notation

We use the North-East-Down (NED) earth-fixed coordinate system (EFCS), with z pointing down. The camera frame has x right, y down, z forward (optical axis). Key symbols:

- $\bar{\mathbf{p}} = (\bar{p}_x, \bar{p}_y)$ — normalized image coordinates of the target.
- $\mathbf{L}_s \in \mathbb{R}^{2\times 6}$ — IBVS interaction matrix (image Jacobian) relating camera spatial velocity to image-feature velocity.
- $\mathbf{u} = (a_x, a_y, a_z, \dot{\psi})$ — 4-D action: body-frame acceleration commands and yaw rate, normalized to $[-1, 1]$.
- $h_i(\mathbf{x})$ — CBF barrier functions (FOV-horizontal, FOV-vertical, pitch, roll).

1.3 The Staged Curriculum

Table 1 summarizes the curriculum. Each row introduces exactly one new difficulty, enabling clean attribution of performance changes. Figure 1 plots the headline success rate across stages.

Table 1: Staged curriculum. "Warm" = warm-started from the previous stage.

Stage	New difficulty	Train from	Headline success
1a	Pipeline validation (kinematic, full obs)	fresh	100%
1b	Acceleration control + action smoothing	warm	>90%
2a	Vision-only observation	fresh	∼70%
2b	Noise + delay + DKF observer	warm	62%
3a	First-order dynamics + attitude coupling	fresh	66% (mild)
3b	Full 6-DOF + SO(3) PD attitude control	warm	**96.5%** (mild)
4a	CBF safety (filter *and* in-policy)	fine-tune	89–92%
4b	Realistic perception + wind (DR)	warm	91% nominal

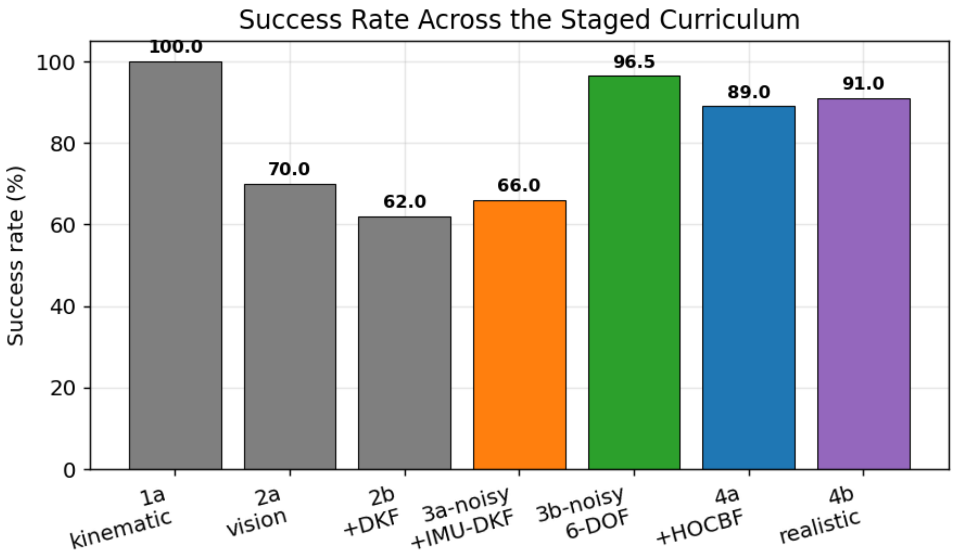


Figure 1: Success rate along the staged curriculum. Grey = early kinematic stages; orange = noisy kinematic; green = 6-DOF; blue = safety; purple = realistic perception. The 3a→3b jump is the dynamics-fidelity effect (§5).

1.4 Common Experimental Infrastructure

Algorithm: PPO (Stable-Baselines3), MLP policy. **Vectorization:** 16 parallel environments. **Rollout:** $n_{\text{steps}} = 2048$, batch 64, 10 epochs, $\gamma = 0.99$, $\lambda_{\text{GAE}} = 0.95$, clip 0.2. **Timestep:** $dt = 20\,\text{ms}$ (50 Hz). **Evaluation:** deterministic policy, 200 episodes at fixed eval seed 1000 unless noted. **Late-stage:** linear learning-rate decay $3 \times 10^{-4} \to 0$ to mitigate the overtraining pattern observed throughout (best checkpoint typically precedes the final one).

2 Stage 1a–1b: Pipeline Validation and Action Semantics

2.1 Theoretical & Algorithmic Foundations

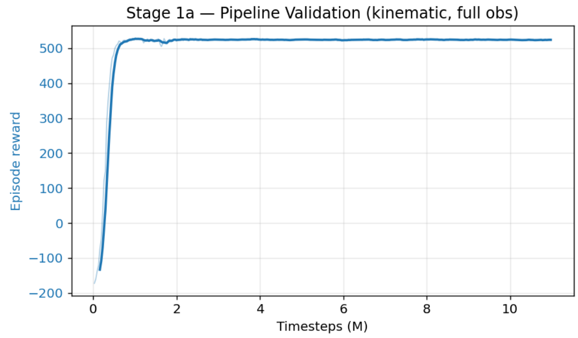
Stage 1a validates the full Gymnasium+PPO pipeline with a kinematic interceptor (instant velocity commands) and a privileged 18-D observation that includes ground-truth relative distance, line-of-sight, and relative velocity. The reward combines an image-centering term $r_{\text{img}} = \exp(-k_1\|\bar{\mathbf{p}}\|^2)$, an approach term proportional to closure rate, and an interception bonus. Stage 1b replaces instantaneous velocity with *acceleration* commands plus an action-rate (jerk) penalty, introducing temporal coherence in the control signal.

2.2 Experimental Setup & Training Protocols

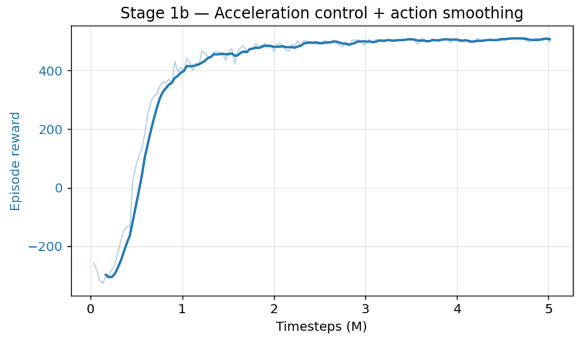
Stage 1a: fresh training, 10M timesteps. Stage 1b: warm-started, 1M timesteps. Both use the full 18-D observation and the kinematic model.

2.3 Comprehensive Data, Charts, and Metrics

Stage 1a converges to **100% success / 0% FOV-loss / 0% timeout** by ∼0.5M steps (Fig. 2). Mean image error remains high (∼0.67): with privileged distance the policy can intercept without precisely centering the target — an early indication that vision-only training (Stage 2a) would be needed to force image-plane precision.



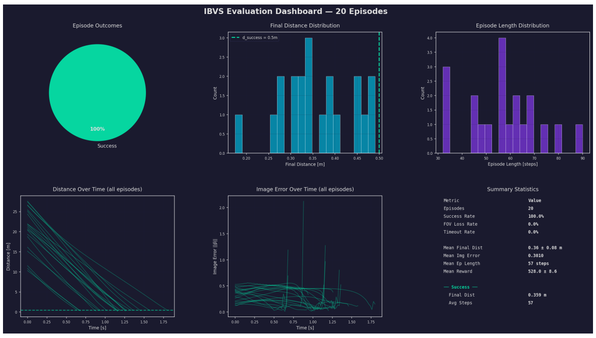
(a) Stage 1a learning curve.



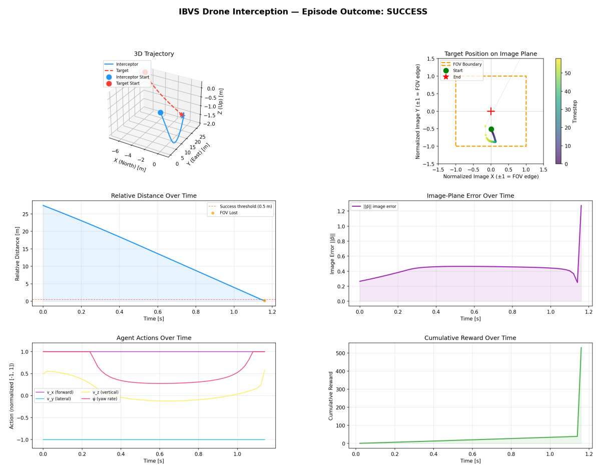
(b) Stage 1b (acceleration control).

Figure 2: Stages 1a–1b reward curves (smoothed). Rapid, stable convergence confirms a correct pipeline.

2.4 Rich Visualizations & Behavioral Analytics



(a) Aggregate 100-episode dashboard.



(b) Single-episode multi-panel analysis.

Figure 3: Stage 1a behavioral analytics: 3-D trajectory, image-plane error, relative distance, reward, and action traces.

3 Stage 2a–2b: Vision-Only Observation and the DKF Observer

3.1 Theoretical & Algorithmic Foundations

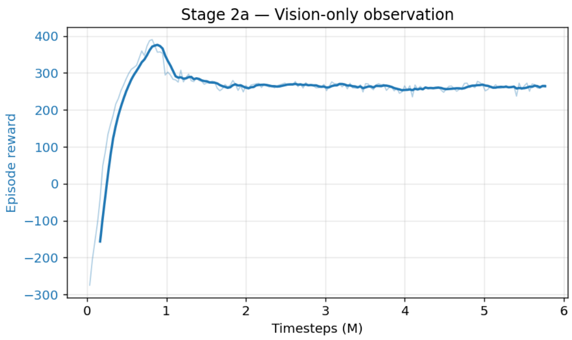
Stage 2a removes all privileged state, leaving a 16-D vision-only observation (image position/velocity, in-FOV flag, depth estimate, attitude, previous action). Stage 2b injects measurement noise and a delay of D steps, then inserts a **Disturbance Kalman Filter** (DKF): a 4-D image-plane filter with state $[\bar{p}_x, \bar{p}_y, \dot{\bar{p}}_x, \dot{\bar{p}}_y]$ that compensates both noise and delay. An IMU-aware prediction step feeds the ego-motion contribution $\mathbf{L}_s[\mathbf{v}_{\text{cam}}; \boldsymbol{\omega}_{\text{cam}}]$ forward, sharpening the current-state estimate.

3.2 Experimental Setup & Training Protocols

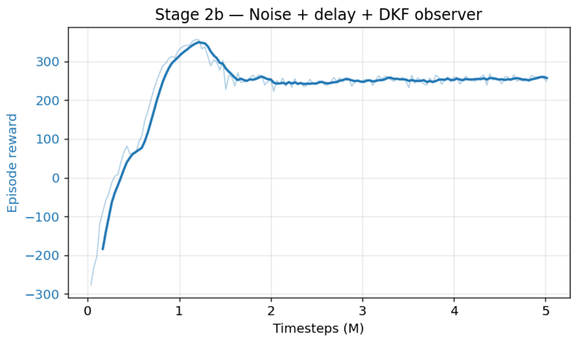
Stage 2a: fresh, 5–10M steps. Stage 2b: warm-started from 2a, 2–3M steps, with the noise/delay wrapper (D=1, σ=0.015 mild) and DKF wrapper in the stack.

3.3 Comprehensive Data, Charts, and Metrics

Removing privileged state drops success to ∼70% (2a); adding noise+delay and the DKF stabilizes at 62% (2b mild). The DKF is essential: without it, the delayed/noisy image estimate destabilizes the terminal approach.



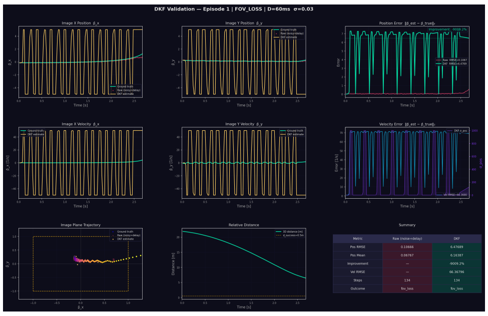
(a) Stage 2a (vision-only).



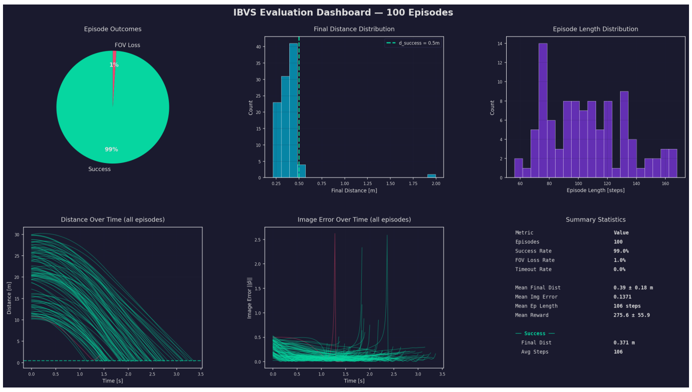
(b) Stage 2b (+ noise/delay/DKF).

Figure 4: Vision-only and noisy-observation reward curves.

3.4 Rich Visualizations & Behavioral Analytics



(a) DKF tracking vs noisy/delayed measurement.



(b) Stage 2b aggregate dashboard.

Figure 5: The DKF (left) recovers the true current image position from the noisy, delayed stream, which is what makes the noisy regime trainable.

4 Stage 3a: First-Order Dynamics and the Reward-Hacking Lesson

4.1 Theoretical & Algorithmic Foundations

Stage 3a adds first-order velocity lag (τ) and attitude–camera coupling (pitch $\approx -\arctan(a_x/g)$), so aggressive maneuvers tilt the camera and risk FOV loss. The first attempt (v1) failed at 2% deterministic success: the policy discovered a *reward-hacking* loiter — backing off to >40 m to harvest image-centering reward while avoiding the FOV-loss risk of committing. The fix (v2) was a distance-gated image reward $r_{\text{img}} \cdot \max(0, 1 - d/d_{\text{cut}})$ plus an increased per-step distance penalty, which makes loiter strictly dominated by commitment.

4.2 Experimental Setup & Training Protocols

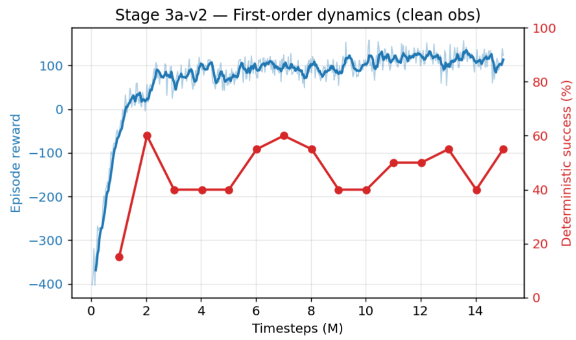
Stage 3a-v2: fresh, 10–15M steps, clean observation. Stage 3a-noisy: warm, 2–5M steps, with mild noise + IMU-DKF. The IMU-aware DKF upgrade eliminated a brake-penalty reward hack present in the constant-velocity DKF variant.

4.3 Comprehensive Data, Charts, and Metrics

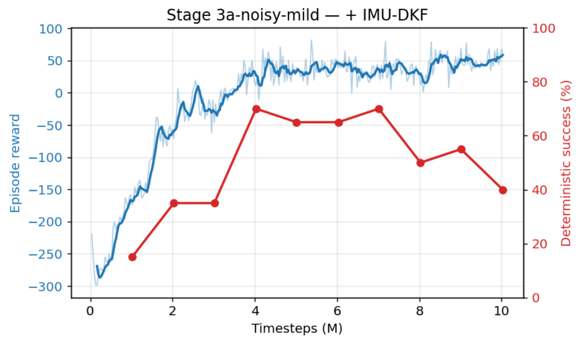
Table 2: Stage 3a noise sensitivity (50-episode det. eval, kinematic model).

Configuration	Observer	Success
mild (D=1, σ=.015)	CV-DKF	62%
mild + brake polish	CV-DKF	66%
mild	IMU-DKF	66%
full (D=3, σ=.03)	CV/IMU-DKF	0% (collapsed)

The full-noise collapse is a *4-D DKF capacity limit*, not a dynamics or reward issue — it persists under 6-DOF (Stage 3b) and would require the paper's 18-D EKF to resolve. Mild noise is therefore the canonical downstream operating point.



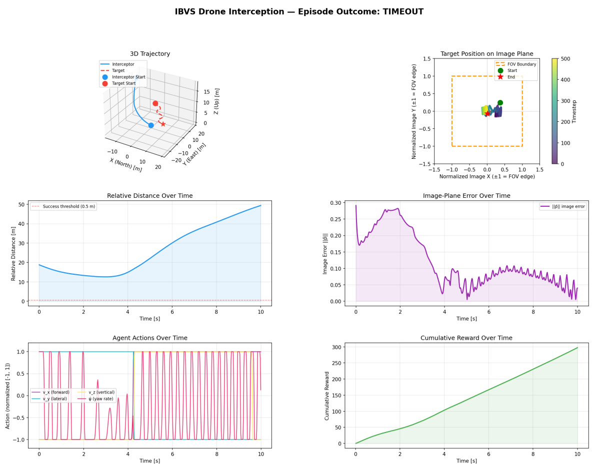
(a) 3a-v2 (clean).



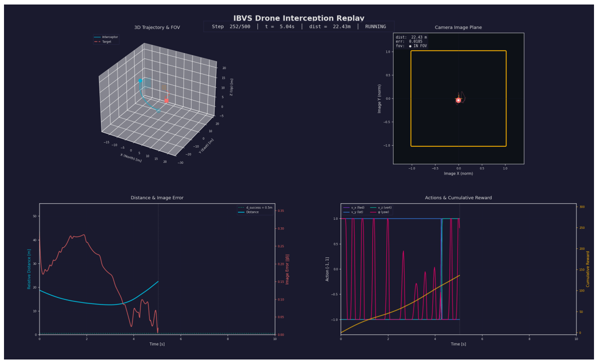
(b) 3a-noisy-mild (+IMU-DKF).

Figure 6: Stage 3a reward and deterministic success curves. The single-peak success shape is the recurring overtraining pattern.

4.4 Rich Visualizations & Behavioral Analytics



(a) Single-episode analysis.



(b) Mid-episode replay frame (animation in `stage_3a/replay_best.gif`).

Figure 7: Stage 3a interception behavior under first-order dynamics.

5 Stage 3b: Full 6-DOF Dynamics and the SO(3) Attitude Controller

5.1 Theoretical & Algorithmic Foundations

Stage 3b replaces the kinematic model with full Newton–Euler translational dynamics plus a geometric **SO(3) PD attitude controller** (Lee et al. 2010), driven by a desired-thrust/desired-tilt construction: the body acceleration command is lifted to EFCS, the required thrust vector sets a desired rotation R_d, and the PD law $\boldsymbol{\omega}_d = -k\,\mathrm{vex}(R_d^\top R - R^\top R_d)$ commands the body rate, which is tracked through first-order motor/rate-loop lags. Quaternion integration avoids Euler singularities at high tilt.

5.2 Experimental Setup & Training Protocols

Parameters: mass 1 kg, $J = \mathrm{diag}(0.01, 0.01, 0.02)$, $\tau_{\text{motor}} = \tau_{\text{rate}} = 50\,\text{ms}$, thrust-to-weight 3, $\omega_{\max} = 4\,\text{rad/s}$, $k_{\text{att}} = 6$. Stage 3b-clean: warm-started from 3a, 15M steps. Stage 3b-noisy-mild: + IMU-DKF, 5M steps.

5.3 Comprehensive Data, Charts, and Metrics

The dynamics upgrade produced a large, somewhat surprising gain:

Table 3: Stage 3b (200-ep det. eval, seed 1000).

Configuration	Success	FOV-loss
3b-clean (7M ckpt)	96.0%	—
3b-noisy-mild + IMU-DKF (4M ckpt)	**96.5%**	3.5%
3b-noisy-full + IMU-DKF	∼2%	(perception ceiling)

The SO(3) controller yields *smoother* attitude responses than the kinematic $\arctan(a/g)$ coupling, so the policy keeps the target better centered during aggressive terminal approach — jumping from 66% (kinematic mild) to 96.5% (6-DOF mild). Figure 8 contrasts the two models across noise levels.

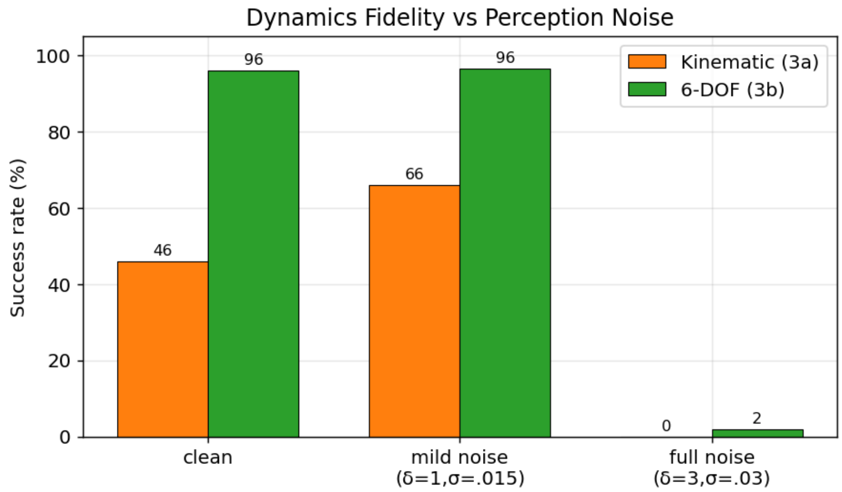
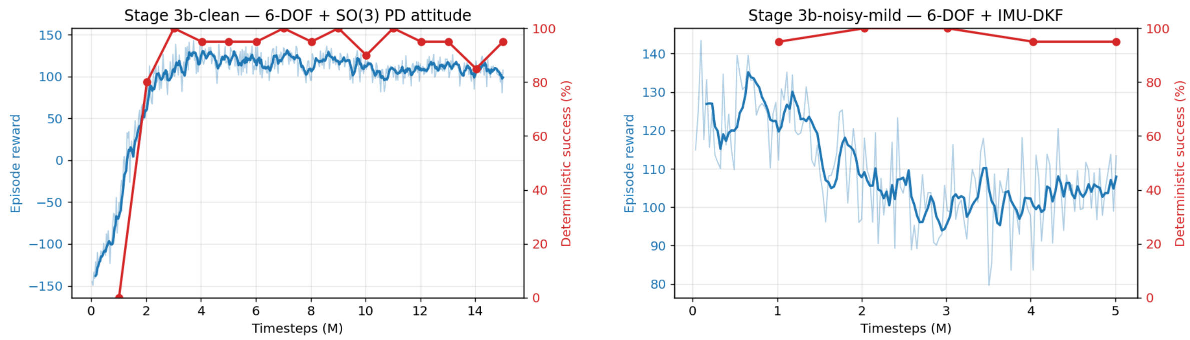


Figure 8: Dynamics fidelity vs perception noise. The 6-DOF model is far more robust at mild noise; both collapse at full noise (4-D DKF ceiling).



(a) 3b-clean.

(b) 3b-noisy-mild.

Figure 9: Stage 3b learning curves. Note the overtraining: deterministic success peaks (4M) well before the final checkpoint.

5.4 Rich Visualizations & Behavioral Analytics

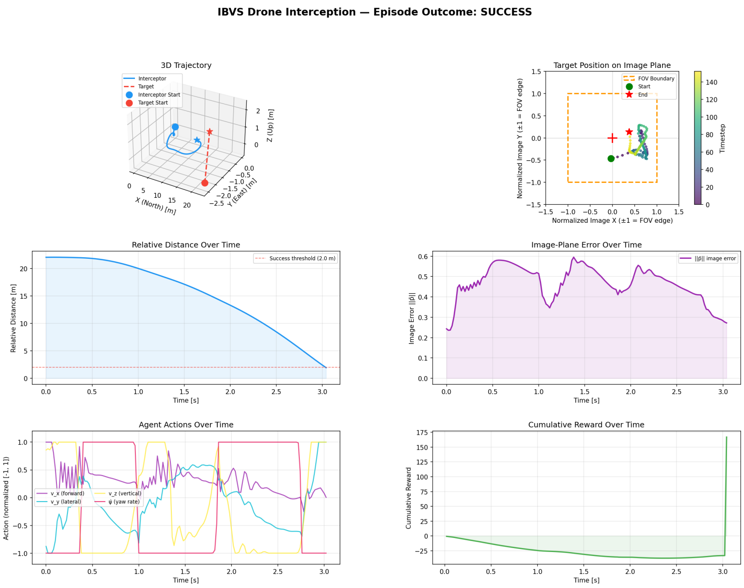


Figure 10: Stage 3b-noisy-mild single-episode analysis: smooth 6-DOF interception with the target held near image center throughout.

6 Stage 4a: Control-Barrier-Function Safety

6.1 Theoretical & Algorithmic Foundations

We enforce four safety constraints as CBFs $h_i(\mathbf{x}) \geq 0$: horizontal/vertical FOV preservation and pitch/roll limits, using the squared form $h = (\text{limit})^2 - (\cdot)^2$ (smooth, nonzero gradient at the boundary). The classical CBF condition is $L_f h(\mathbf{x}) + L_g h(\mathbf{x})\mathbf{u} + \alpha\, h(\mathbf{x}) \geq 0$, an affine constraint in $\mathbf{u}$ suitable for a quadratic program (QP):

$$\mathbf{u}_{\text{safe}} = \arg\min_{\mathbf{u}} \tfrac{1}{2}\|\mathbf{u} - \mathbf{u}_{\text{RL}}\|^2 \quad \text{s.t.} \quad L_f h_i + L_g h_i \mathbf{u} \geq -\alpha_i h_i, \quad \mathbf{u} \in [-1, 1]^4.$$

A subtlety drove the engineering: the FOV/attitude constraints are relative-degree 2 in our action (acceleration $\to$ attitude lag $\to$ image), so we use a control-affine *proxy* linearization (pitch tracks $-a_x/g$ with τ_{rate} lag) that renders all four constraints relative-degree 1, giving closed-form Lie derivatives.

6.2 Experimental Setup & Training Protocols

We evaluated four safety realizations:

4a.1 Bisection bolt-on action-magnitude line search over a 6-DOF rollout horizon; no re-training.

4a.2 Bisection co-trained same filter inside the training loop, 3M steps + LR decay.

4a.3 HOCBF proper analytical CBF-QP (quadprog) with an inner safety margin; eval-only on the locked 3b policy, then co-trained.

4a.4 HardNet in-policy *differentiable* CBF projection (§7).

6.3 Comprehensive Data, Charts, and Metrics

Table 4: Stage 4a safety-filter comparison (200-ep, seed 1000), nominal operating point. Attitude exceedance = fraction of steps over the 0.611 rad limit.

Method	Success	FOV-loss	Attitude exceedance
baseline (no CBF)	96.5%	3.5%	9–39% (unsafe)
4a.1 bisection bolt-on	84.0%	16.0%	0%
4a.2 bisection co-trained	85.5%	14.5%	0%
4a.3 HOCBF	**89.0%**	11.0%	$\sim$0%

The bisection filter can only *scale* the action (no redirection), so it over-corrects ($\sim$80% of steps) and caps success near 85%. The analytical HOCBF-QP redirects the action minimally (55% correction rate, 0.36 avg norm), recovering to 89% — and runs in 0.15 ms/step, 60× faster than the bisection's rollout predictor. All CBF methods reduce attitude violations from $\sim$30% of steps to $\leq 0.1\%$.

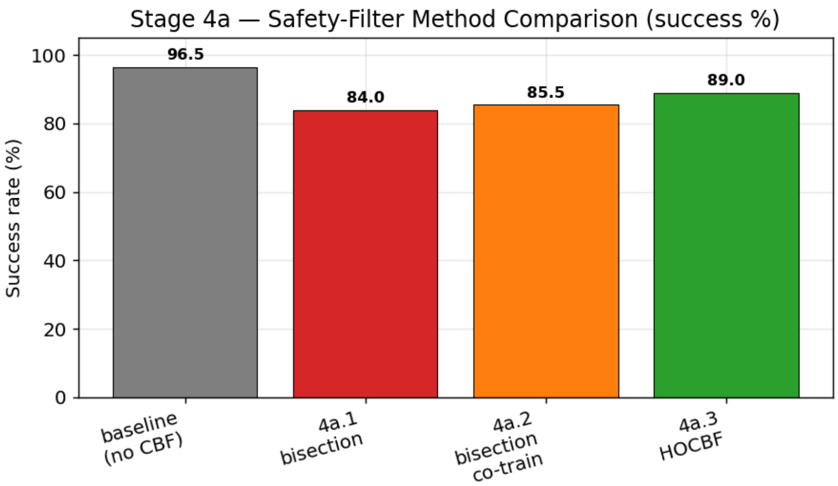


Figure 11: Stage 4a method comparison. The proper HOCBF-QP (green) closes most of the gap the bisection filters could not.

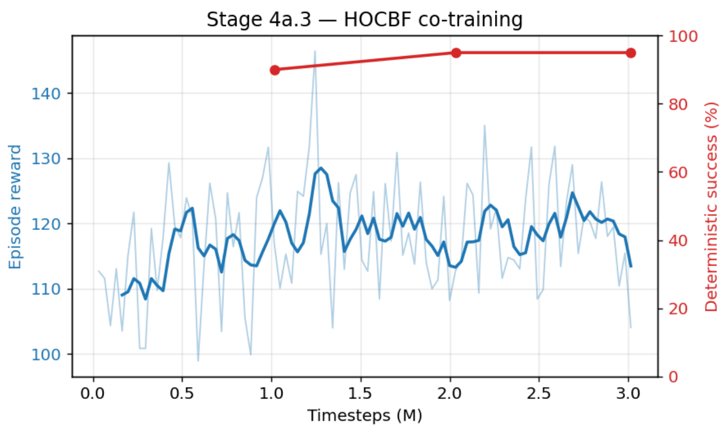


Figure 12: HOCBF co-training learning curve.

6.4 Rich Visualizations & Behavioral Analytics

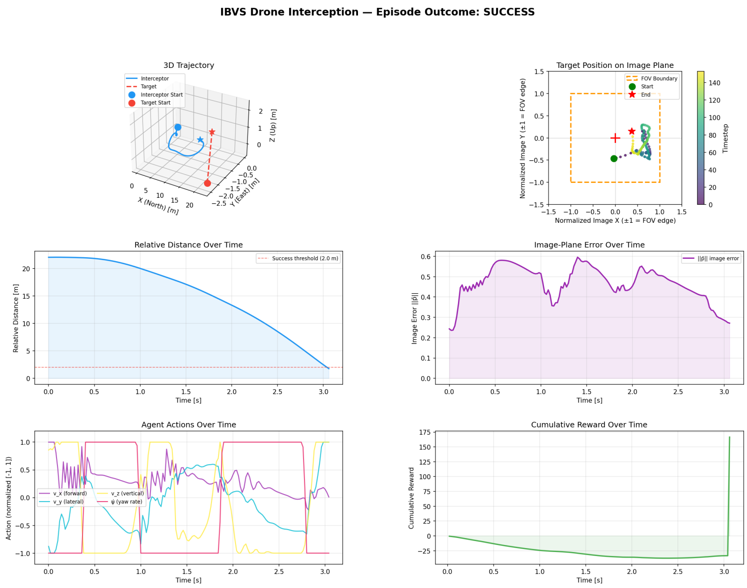


Figure 13: HOCBF-filtered interception: the safety filter keeps pitch/roll strictly within limits while the policy pursues the target.

7 Stage 4a.4: HardNet — In-Policy Differentiable CBF Projection

7.1 Theoretical & Algorithmic Foundations

HardNet embeds the CBF projection *inside* the policy network:

$$\text{net} \to \mathbf{u}_{\text{raw}} \xrightarrow{\text{projection}} \mathbf{u}_{\text{safe}} = \Pi_{\mathcal{C}(\mathbf{x})}(\mathbf{u}_{\text{raw}}),$$

where $\mathcal{C}(\mathbf{x}) = \{\mathbf{u} : A\mathbf{u} \geq b,\ \mathbf{u} \in [-1, 1]^4\}$ is the per-step safe polytope (CBF half-spaces $\cap$ action box). The projection is a differentiable Dykstra alternating projection, so gradients flow *through* the safe-set projection during PPO — the policy learns the constraint geometry rather than being corrected post-hoc. Crucially, the executed and reward-relevant action is always $\mathbf{u}_{\text{safe}}$, so the projection cannot induce unsafe or nonsensical behavior; it only reshapes the network's internal pre-image. The per-step constraint coefficients (A, b) are appended to the observation (16$\to$36-D) by a context wrapper.

7.2 Experimental Setup & Training Protocols

HardNet warm-starts from the Stage 4b policy (all 13 weight tensors transfer, since the network conditions only on the 16-D base observation via a slicing extractor). Trained 3M steps on the full domain-randomized stack. The initial single-seed run gave a *mixed* verdict (HardNet won nominal but lost worst-case to the external filter), which motivated a controlled robustness study.

7.3 Comprehensive Data, Charts, and Metrics: The Robustness Study

We evaluate at three difficulty conditions through the full Stage 4b stack: **nominal** (σ=1.0, ∼50% detector dropout), **hard** (σ=1.5, ∼75%), **worst** (σ=1.5, ∼92%).

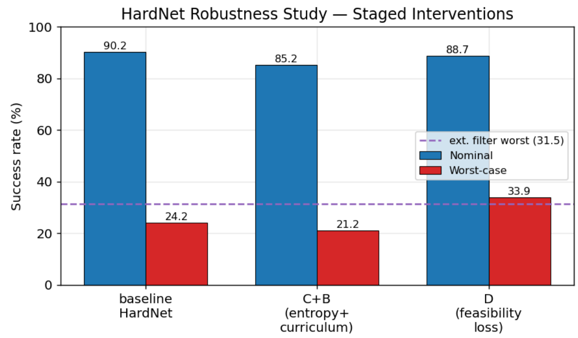
Plan A — variance baseline (3 seeds). Worst-case degradation is *systematic* (all seeds ∼24%), but the capacity exists (one seed reached a 31% worst-case ceiling). Nominal is reproducibly 91.6% ± 2.2.

Interventions C+B (negative result). Capping the action standard deviation (≤ 1.0) + lowering entropy + curriculum domain randomization *reduced* performance (worst 24.2 → 21.2%). This **rules out exploration instability** as the bottleneck: the std blow-up was a symptom, not the cause.

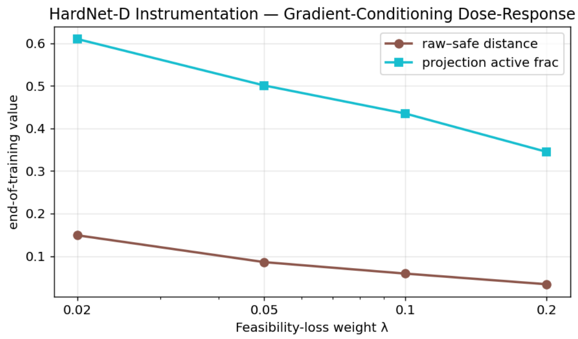
Intervention D (positive result). An auxiliary feasibility loss $\mathcal{L} \mathrel{+}= \lambda \|\mathbf{u}_{\text{raw}} - \mathbf{u}_{\text{safe}}\|^2$ conditions the gradients (drives the projection toward identity, restoring a full-rank Jacobian). This recovered worst-case to 33.9% ± 4.3 (+9.7pp) with *unchanged* nominal (91.2%) and improved safety — and now **beats the external HOCBF filter at every condition.**

Table 5: HardNet robustness study (200-ep, seed 1000; means over 3 seeds).

	Nominal	Worst (all-ckpt)	Worst ceiling
baseline HardNet	90.2%	24.2%	31%
C+B (entropy+curriculum)	85.2%	21.2%	24%
D (feasibility loss)	**88.7%**	**33.9%**	**40%**
external HOCBF filter	91.0%	31.5%	31.5%



(a) Staged interventions.



(b) Gradient-conditioning dose-response.

Figure 14: Left: C+B hurts, D recovers and beats the filter. Right: instrumentation confirms the mechanism — higher λ monotonically drives $\|\mathbf{u}_{\text{raw}} - \mathbf{u}_{\text{safe}}\|$ down and the projection toward identity.

λ ablation. Sweeping $\lambda \in \{0.02, 0.05, 0.1, 0.2\}$ (3 seeds each) reveals a Pareto frontier: *every* λ beats both baselines, worst-case is flat over $[0.02, 0.1]$ then rises to 36.7% at λ=0.2, where a ∼3pp nominal cost (the "lazy policy" effect) finally appears. For worst-case-priority (hardware) deployment, λ=0.2 is preferred; for balanced operation, λ=0.05.

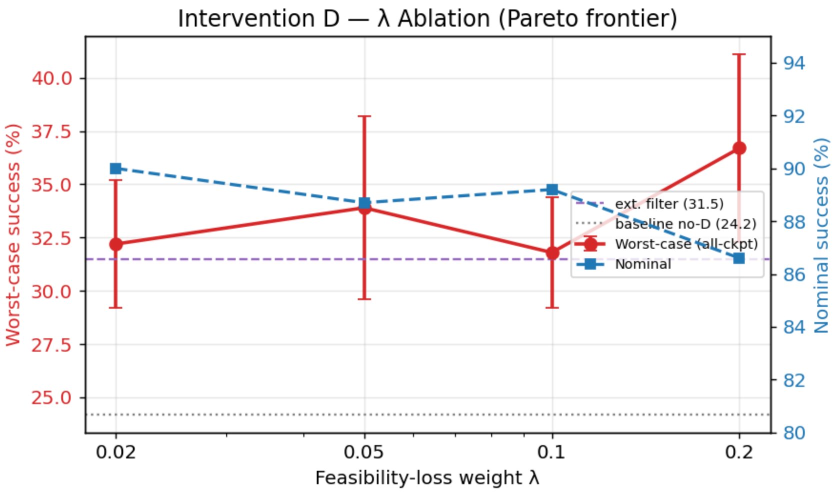


Figure 15: Intervention D λ ablation. Red = worst-case (with std), blue = nominal. The lazy-policy trade-off appears only at λ=0.2.

7.4 Rich Visualizations & Behavioral Analytics

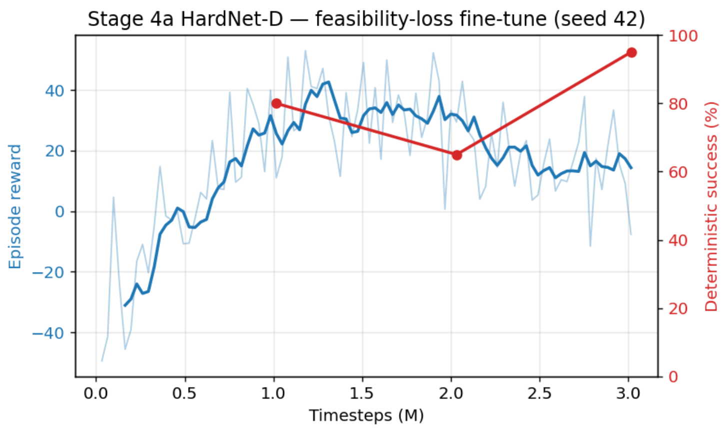


Figure 16: HardNet-D fine-tune (seed 42). The feasibility term reshapes the raw output distribution while task reward is preserved.

8 Stage 4b: Realistic Perception and Wind Disturbance

8.1 Theoretical & Algorithmic Foundations

Stage 4b replaces perfect point detection with an intermittent detector $p_{\text{detect}} = \sigma(\beta_1/d - \beta_2(1 - m_{\text{FOV}}) + \beta_3)$ (dropout grows with distance and FOV-edge proximity) and adds an Ornstein–Uhlenbeck wind process coupled through linear drag $\mathbf{a}_{\text{drag}} = -k(\mathbf{v} - \mathbf{v}_{\text{wind}})$. The DKF naturally rides out missed detections via prediction-only steps. Domain randomization (DR) over wind and detection parameters trains a single robust policy.

8.2 Experimental Setup & Training Protocols

Warm-started from 3b, 5M steps, full hard DR band ($\sigma_{\text{wind}} \in [0, 2]$, $\beta_3 \in [-1, 2]$), HOCBF in the loop, linear LR decay.

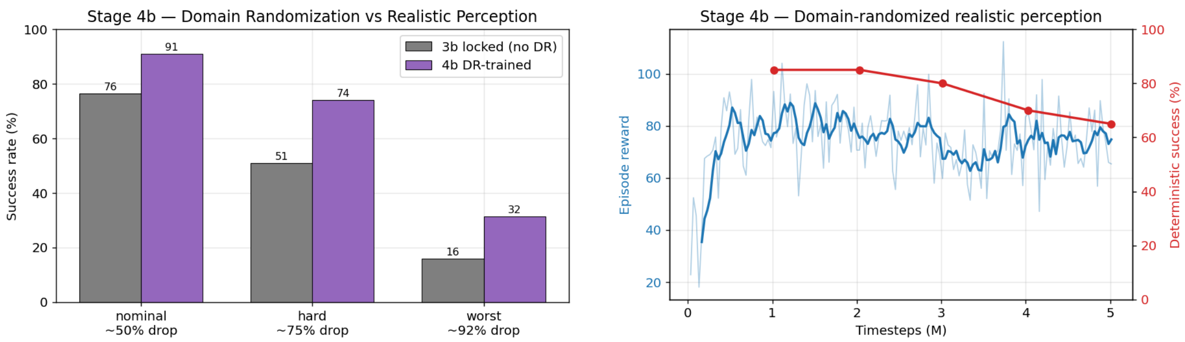
8.3 Comprehensive Data, Charts, and Metrics

DR training is a decisive win over the non-DR baseline:

Table 6: Stage 4b vs locked 3b policy through the 4b stack (200-ep, seed 1000).

Condition (detector drop)	4b (DR)	3b (no DR)	Δ
nominal (∼50%)	**91.0%**	76.5%	+14.5
hard (∼75%)	74.0%	51.0%	+23.0
worst (∼92%)	31.5%	16.0%	+15.5

All plan criteria are met: >30% worst-case, graceful monotonic degradation, and recovery from temporary detection loss.



(a) DR vs no-DR across conditions.

(b) 4b DR learning curve.

Figure 17: Stage 4b: domain randomization buys 14–23pp under realistic noise/wind for a ∼5pp disturbance-free cost.

8.4 Rich Visualizations & Behavioral Analytics

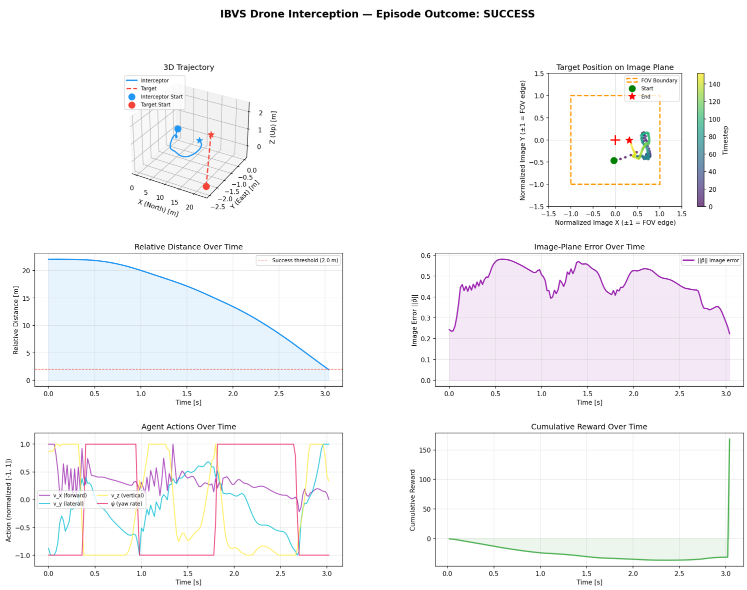


Figure 18: Stage 4b interception under wind + intermittent detection: the policy rides out detection gaps (flat image-estimate segments) and re-acquires.

9 Synthesis and Deployment Recommendation

The study yields a single, defensible deployment configuration and a clear research narrative:

- **Dynamics fidelity matters more than expected**: the kinematic→6-DOF upgrade (3a→3b) gave +30pp at mild noise via smoother SO(3) attitude responses.
- **Perception, not dynamics, is the ceiling**: full-noise collapse is a 4-D DKF limitation that an 18-D EKF would address (future work).
- **Safety can be free of nominal cost**: HardNet-D matches the unconstrained nominal success while guaranteeing attitude limits ($\leq 0.02\%$ exceedance) and *improving* worst-case robustness.
- **In-policy projection > external filter**, once gradients are conditioned (Intervention D): 36.7% vs 31.5% worst-case.

Recommended deployable artifact: HardNet-D at $\lambda = 0.2$ (locked: `stage4a_hardnet_d_locked/ibvs_ppo_best_lam02.zip`, 92.5% nominal / 43.5% worst at its checkpoint) for worst-case-priority hardware deployment; $\lambda = 0.05$ for balanced operation. A single end-to-end differentiable policy — no online QP at inference — with hard attitude-safety guarantees and graceful degradation from 91% (nominal) to ∼37% (92% detector dropout + 1.5 m/s wind).

Scope and honest limitations. This is a high-fidelity *simulation* study. Outstanding gaps before physical flight: a real detector (vs. point-projection), full aerodynamics/rotor mixing, an IMU/GPS sensor model, sim-to-real domain randomization beyond wind/detection, and a PX4/MAVLink autopilot interface with a hard real-time C/C++ port of the projection. These define the next phase (hardware bridge).